# FedCode: Communication-Efficient Federated Learning via Transferring Codebooks

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## 01 Context & approach

### The challenge

Clustering model weights reduces value storage, but transmitting every weight index still costs bandwidth. *FedCode* usually exchanges only cluster centers in both directions. Periodic calibration also transfers compressed assignments to realign client and server models.

### Our contribution

Reduce upstream and downstream traffic by exchanging compact codebooks.
Map existing weights to received centers without sending every index each round.
Use periodic calibration to control the communication-accuracy trade-off.

### 1 Broadcast centers

Cluster global weights and send sorted centers. Clients map existing weights to their nearest received centers.

### 2 Train and return codebooks

Clients train and cluster the full model. The server concatenates received codebooks and maps its weights to the nearest centers.

### 3 Calibrate periodically

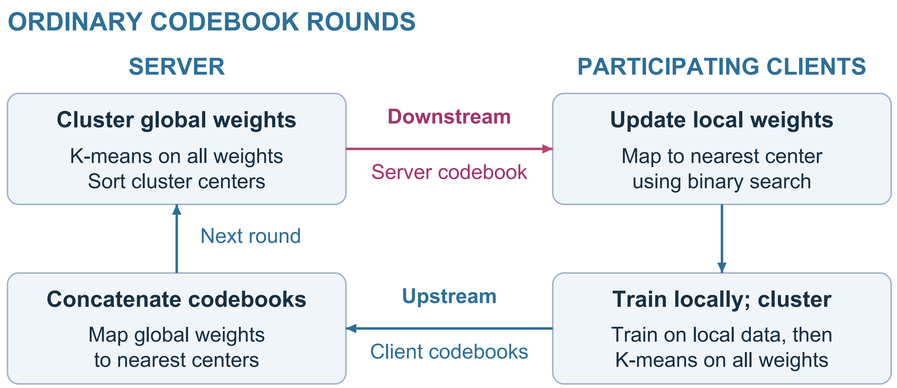
Also transfer compressed weight indices on calibration rounds. Reconstruct models; server-side calibration then applies FedAvg.

### Scope

Communication volume is measured; client latency, energy and run-to-run uncertainty are not established. Calibration direction labels conflict in the paper; F1/F2 are retained without assigning directions.

## 02 Method

**Exchange codebooks in both directions and periodically calibrate weight assignments.**




Codebook = cluster centers; ordinary rounds omit weight indices.

**PERIODIC CALIBRATION — SELECTED ROUNDS**

Transfer codebook + weight indices in the relevant direction.

Clients reconstruct weights from indices and centers.

The server reconstructs client models, then applies FedAvg.

### Core formulation

$$j^* = \arg\min_j |w_i - c_j|, \qquad w_i \leftarrow c_{j^*}$$

$w_i$: existing weight; $c_j$: received center. Ordinary rounds choose the nearest center; the server uses concatenated client codebooks.

$$\mathrm{DTR} = \frac{2Pb}{2Kb + (F_1 + F_2)P\lceil \log_2 K \rceil}$$

P weights, K centers, b bits/weight; F1 and F2 are calibration frequencies. Less index transfer reduces traffic.

Paper Code

## 03 Results

**14.6×**
**joint communication reduction**
IID CIFAR-100; ResNet-20.

**58.07%**
**FedCode test accuracy**
FedAvg: 61.22% on the same workload.

### Compress both communication directions

| Method | Accuracy (%) | Down DTR | Up DTR | Joint DTR |
|---|---|---|---|---|
| FedAvg | 61.22 | — | — | — |
| OMC-T | 61.21 | — | 2.9 | 1.4 |
| FedCompress | 59.67 | 4.11 | 4.11 | 4.11 |
| FedZip | 57.98 | — | 19.8 | 1.9 |
| **FedCode** (Ours) | **58.07** | **24.2** | **10.4** | **14.6** |

IID CIFAR-100, ResNet-20; 10 clients, full participation, 100 rounds, 4 local epochs. *FedCode*: 64 centers, F1=0.2, F2=0.5. DTR = FedAvg bits / method bits; higher is better.

### Calibration controls the trade-off

| F1 / F2 | Accuracy Δ (pp) | Joint DTR | Data (MB) |
|---|---|---|---|
| 1 / 1 | -0.86 | 5.3 | 394.21 |
| 0.2 / 0.5 | -3.15 | 14.6 | 143.92 |
| 0.1 / 0.1 | -21.64 | 44.3 | 47.35 |
| 0 / 0 | -46.88 | 264.7 | 7.9 |

Same CIFAR-100/ResNet-20 setting; 64 centers. Accuracy change is relative to FedAvg; negative means lower accuracy. FedAvg communication: 2102 MB.

### Takeaway

**FedCode** achieves **14.6× less joint communication** at 58.07% accuracy, 3.15 percentage points below FedAvg in this setting. Without calibration, communication falls further but accuracy drops 46.88 points below FedAvg.